Mingyang (Yang) Liu

PhD in Finance, Imperial College London · Empirical asset pricing and financial econometrics

Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Imperial College London

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Legal name: Yang Liu (刘杨). Mingyang (刘明杨) is the name I adopted on coming of age; some listings use Yang Liu.

Research Summary

- **Research programme:** my papers study the ruler, the metric a characteristic-based factor model uses to measure and compare directions in characteristic space. The ruler adds no information, yet it can decide which factors a model identifies, how fitted returns are split between factors and alpha, and which portfolio an investor holds. Central paper: The Ruler Matters.
- **Methods:** characteristic-based and latent-factor models (IPCA and No-Arbitrage IPCA) on large U.S. stock-characteristic panels; identification and inference; rolling out-of-sample portfolio evaluation. Author of `naipca`, a Python toolkit for IPCA-family factor estimation.
- **Teaching:** Graduate Teaching Assistant in Financial Statistics (student rating 4.8 out of 5; nominated twice for the Excellence in Teaching Award); mentor at the Imperial Student Investment Fund, which works as a student finance lab.
- **Forecast evaluation:** built a workflow that compares forecasting models on the same forecast origins, tests whether differences in forecast accuracy are statistically significant (Diebold–Mariano) and shows how far a model ranking depends on the evaluation window and output version (archived UK services-inflation example, 3- and 6-month horizons, pseudo-out-of-sample design with look-ahead documented).

Research Fields

Empirical asset pricing (cross-sectional return prediction, characteristic-based factor models); machine learning and big data in finance (regularised and latent-factor estimators on large characteristic panels); financial econometrics (identification and inference in high-dimensional factor models); portfolio choice and risk management.

Education

Aug 2019–Aug 2026 — **Imperial College London**, Imperial Business School, Department of Finance, London
PhD in Finance, conferred August 2026
Supervisors: Prof. Paolo Zaffaroni (main) and Prof. Pasquale Della Corte
Thesis: No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing. Examined March 2026.
MRes in Finance, with Distinction, 2020

Aug 2017–Feb 2019 — **Columbia University**, Graduate School of Arts and Sciences, New York
M.A. in Economics (Financial Econometrics; STEM-designated programme)
Master's thesis: Diffusion Index Forecasting: A Robust Estimator for Common Shock, supervised by Prof. Jushan Bai

Aug 2014–Apr 2017 — **University of Michigan**, Ann Arbor
B.S., double major in Economics and Mathematics (Mathematics of Finance and Risk Management track)

Research

Working Papers

JOB MARKET PAPER **The Ruler Matters: Factor Exposures, Attribution, and Portfolio Risk**

Working paper, October 2026

Seminar and conference presentations: University of Cambridge, Faculty of Economics, 4 November 2026 (invited; scheduled); EPFL, Lausanne, 16 June 2026 (invited seminar; precursor presented as No-Arbitrage Factor Geometry)

Uses the cross-sectional second moment (Gram matrix) of characteristics as a ruler for IPCA loading spaces. Expanding from 36 to 153 characteristics can raise in-sample fit yet leave momentum and size positions less fully represented; reallocating shared exposures changes attribution, not joint fitted returns. With coefficients fixed, exposure references change holdings, trading and subsequent risk.

PDF

Characteristic-Space Metrics in Factor Models: Identification and Inference

SSRN preprint, September 2026. DOI: 10.2139/ssrn.7445120

Factor models with identical fit can credit returns differently to latent factors, observed factors and an intercept; fit cannot choose the split, but the inner product chosen for characteristic space does. The paper derives rotation-invariant inference that carries the Gram metric's estimation error, and drift-robust confidence intervals near an intercept cap (exposure budget).

PDF · Online supplement · SSRN

A Geometric Framework for Identification in Characteristic-Based Factor Models

SSRN preprint, March 2026 (preliminary version). DOI: 10.2139/ssrn.7013178. Grows out of the PhD thesis; its inference theory is developed further in Characteristic-Space Metrics.

Using a metric taken from the long-run cross-sectional distribution of firm characteristics, the paper derives NA-IPCA, which nests IPCA, caps how much the intercept may carry and admits rotation-invariant inference. In the headline calibrated simulation, NA-IPCA recovers the latent pricing span and an intercept-free IPCA benchmark does not; that design does not isolate the metric.

PDF · SSRN

Same Forecasts, Different Portfolios: Factor Normalization and Regularization

Working paper, October 2026

Two IPCA fits under different normalisations give nearly identical forecasts but different gross Sharpe ratios: a ridge added to factor covariances penalises exposures in units the normalisation sets. Writing the penalty consistently aligns the portfolios. Using the full fitted mean, intercept included, raises the gross Sharpe ratio; its net advantage since October 2007 is imprecise.

PDF

Interpreting Estimated Pricing Errors: Evidence from Characteristic-Based Return Forecasts

Working paper, September 2026

What does a smaller estimated pricing error (alpha) tell a model user? On 153 U.S. characteristics, crossing the expected-return estimates and factor spaces of one model's equal- and value-weighted fits shows that their gap in fitted intercept magnitudes mainly reflects different estimated means; residual diagnostics add little to forecast selection (exploratory evidence).

PDF

Research in Progress

Exposure or Intercept? Identifying Corporate Bond Premia with No-Arbitrage IPCA

with Paolo Zaffaroni · in preparation

Asks, with No-Arbitrage IPCA on public TRACE data, whether bond characteristics predict returns through factor exposure or an intercept, and how much of that split the choice of ruler (the metric) and the intercept cap decide.

PhD Thesis

No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing

PhD thesis, Imperial College London; degree conferred August 2026. Thesis available on request.

Extends IPCA, via the empirical characteristic metric, to nested QZ-IPCA and NA-IPCA models separating latent, observable (Fama–French five factors plus momentum) and residual-pricing components under an intercept cap; gives identification results, an asymptotic linear expansion and projected alternating least squares; simulations and a Jensen–Kelly–Pedersen U.S. panel show near-identical fit coexisting with different identified pricing directions.

Positions

Jun 2026–present

Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Department of Finance, Imperial Business School, Imperial College London

Host: Prof. Robert Kosowski, Co-Director of the Centre

Research agenda for the visit: economically constrained machine learning for cross-sectional asset pricing and portfolio choice.

- Machine learning and high-dimensional data: extending the No-Arbitrage IPCA framework of the thesis to regime-switching factor models in which risk premia vary with macro conditions; combining penalised estimation with economically constrained representation learning to separate alpha signals that survive those constraints from spurious signals driven by weak factors or data snooping.

Research Experience

Aug 2019–Aug 2026 **Doctoral research**, Department of Finance, Imperial College London (asset pricing, financial econometrics, machine learning in finance)

- Models and geometry: characteristic-based factor models with an explicit characteristic-space geometry, using the empirical Gram metric of firm characteristics as the identification convention; the extension of IPCA to the nested QZ-IPCA and NA-IPCA models developed in the thesis.
- Identification and inference: rotation-invariant identification (loading classes modulo latent rotations); asymptotic linear representations that carry the metric's estimation error into standard errors; drift-robust confidence intervals near an intercept cap (exposure budget); projected alternating least-squares estimation; geometry-calibrated Monte Carlo designs.
- Portfolio experiments and trading costs: rolling out-of-sample portfolios (120-month windows, U.S. stocks 1973–2024) that trace weight gaps across factor normalisations to the units of a factor-covariance ridge, with returns after proportional trading costs; IPCA models with libraries of 36 and 153 characteristics compared by the stock positions their loading spaces represent, with value, momentum and size hedges traced to holdings, trading and subsequent risk; forecast-evaluation designs that cross expected-return estimates with factor spaces.
- Data and tools: the Jensen–Kelly–Pedersen library of 153 U.S. stock characteristics (jkpfactors.com; WRDS `contrib.global_factor`, built from CRSP/Compustat), 1962–2025, with point-in-time formation alignment; a CRSP/Compustat panel built following the Kelly–Pruitt–Su replication dataset (36 characteristics plus a constant, 12,813 firms, 1964–2014); CRSP daily dollar volume for trading-capacity checks; NBER recession dates. Author of `naipca`, a Python toolkit implementing IPCA, QZ-IPCA and NA-IPCA with a companion metric-diagnostics package (available to academic users on request).

2026 **Research Toolkit demo** (independent research; archived example): built the forecast-evaluation workflow shown in Macro Forecast Evaluation & Regime Diagnostics.

May 2018–May 2019 **Research Assistant (macro forecasting)**, Department of Economics, Columbia University

- Diffusion-index and factor models that extract common inflation and interest-rate factors from large mixed-frequency macro data; robust PCA and stable principal component pursuit for missing data, revisions, outliers and structural breaks.

Jun 2016–Jan 2017 **Research Assistant (derivatives pricing)**, Department of Mathematics, University of Michigan

- Real-option valuation of a gold mine, using the Lihir Gold Limited IPO as a case: estimated gold-price volatility and the present value of total costs, took strike parameters from the prospectus, and priced the option with a binomial tree.

Research Projects

Jul 2026–present **Ongoing research collaboration** with Prof. Semyon Malamud (EPFL and Swiss Finance Institute) on portfolio construction from linear out-of-sample return forecasts. The project complements his work on nonlinear return prediction and began after a seminar presented at EPFL in June 2026 at his invitation.

Teaching and Mentoring

2023, 2024 **Graduate Teaching Assistant**, Imperial Business School

Financial Statistics, MSc Risk Management & Financial Engineering. Student rating 4.8 out of 5.

Nominated by students for the Excellence in Teaching Award (both years).

Teaching notes (PDF, 96 pp., 15 MB, shared for academic use):
Teaching notes: Financial Statistics

May 2024–present **Imperial Student Investment Fund, a student finance lab** · Adviser; former Chair of the Executive Committee and Advisory Committee

- The fund works as a student finance lab in which students apply what they learn in class to real markets. Working with the Head of the Department of Finance, helped develop it as a teaching lab and build its Python research platform on real trade-level market data from the LSEG database.
- Mentored students in turning course theory into investment decisions, and led the fund's student teams in the 2026 QRT Quantitative Competition, a paper-trading competition (four teams reached the final, two placed second and third).

- Organised and coordinated the lab's teaching activities with Qube Research & Technologies (QRT) and LSEG, including QRT's Trading and Risk Management training sessions and portfolio-optimisation training on LSEG Workspace.

Scholarships and Honours

2019–2025	Fully funded doctoral scholarship, Imperial Business School (Graduate Teaching Assistant Scholarship)
2014–2016	University Honors, University of Michigan

Skills and Languages

Programming	Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, PyTorch), MATLAB, R, SQL, Git
Methods	Cross-sectional asset pricing (IPCA, QZ-IPCA, NA-IPCA, Fama–MacBeth regressions), latent-factor models, regularised regression and portfolio shrinkage (ridge-type rules, UPSA), robust PCA (RPCA, SPCP), moving-block bootstrap inference, Monte Carlo simulation design, neural networks and transfer learning, state-space and regime-switching models, financial text analysis
Data	CRSP/Compustat and the Jensen–Kelly–Pedersen characteristic library (jkpfactors.com) via WRDS; Bloomberg; LSEG Workspace
Languages	Chinese (Mandarin, native); English (fluent; working language)

References

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Professor Robert Kosowski
Host, visiting researcher appointment
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